

LiSA: Light-Space Atlases for Relightable Gaussian Splatting

Anonymous CVPR submission

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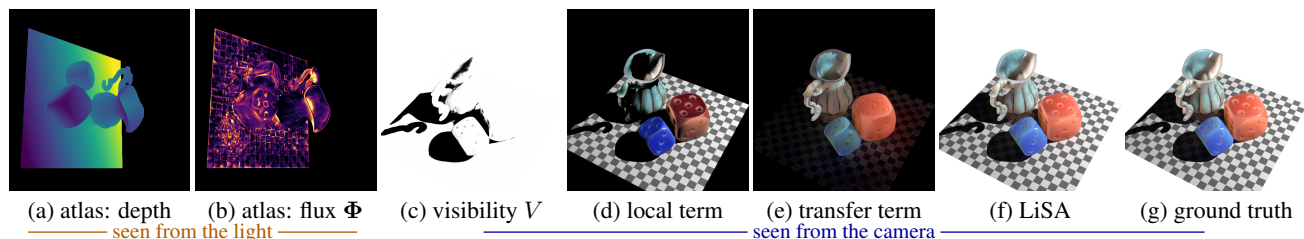


Figure 1. **Relighting from the light’s point of view.** One rasterization of the Gaussians from the point light records how deep the light is absorbed (a) and what it hits, through learned flux features (b, one of eight channels). A camera pixel reads this atlas to decide whether it is in shadow (c) and how much light it reflects locally (d); the transfer term (e) adds radiance that the model assigns mostly to the translucent vase and dice, a learned split rather than a physical decomposition. The sum (f) matches the test view (g) at 34.4 dB, against 32.6 dB for the best of four recent baselines. Components (d, e) add in linear radiance and are shown on black.

Abstract

When a point light moves around an object, much of the change in its appearance comes from where the light lands. Relightable Gaussian splatting methods estimate this one primitive at a time; we instead reason in the light’s image plane. When the Gaussians are rasterized from the light, the alpha-compositing weights are the fractions of each light pixel’s flux that the primitives intercept. LiSA stores this deposit in a light-space atlas of depth moments and learned flux features. Any receiver, a Gaussian during training or a pixel at render time, reads a pyramid of the atlas to obtain a filtered shadow test refined by a small network and a transfer term linear in the deposited features. Training also required computing the loss on linear rather than display-encoded radiance while the geometry forms, which raises held-out PSNR by 6–7 dB in a controlled study on two scenes with thin, cluttered structures. On 18 scenes of three public benchmarks, LiSA reaches 33.47 dB average PSNR against 31.59 dB for the strongest of four recent baselines, is the only method among the top two on every benchmark, and trains in about 20 minutes. Our ablations also delimit the transfer term: it matters most on translucent materials, whereas a local network of equal size matches it on most others.

1. Introduction

Photographing an object one light at a time (OLAT) and relighting it from new viewpoints is one of the oldest routes to photorealistic digital assets [13]. Its modern form reconstructs a relightable representation from hundreds to thousands of images, each lit by a single calibrated point light [3, 62], and Gaussian splatting [31] has made the result renderable in real time [5, 17, 33, 35, 66].

Under a point light, much of what makes an image look right is settled before any material is evaluated, namely where the light lands and where it does not. Current Gaussian methods settle this one primitive at a time. GS³ [5] and SSD-GS [66] splat the scene toward the light to give every Gaussian one visibility value, RNG [17] reduces a depth shadow map to a scalar cue, and translucency is predicted by per-Gaussian networks from local inputs [15, 66]. A large Gaussian on a floor then holds a single shadow value, so every moving shadow boundary has to be carved out by densification.

We instead reason in the image plane of the light. When the Gaussians are rasterized from a point light, the alpha-compositing weights split the flux of every light pixel among the primitives along its ray; within the splatting model, they are the fractions of that flux each Gaussian intercepts. One extra rasterization thus records how much light each primi-

tive absorbs and at which depth, which we call the *deposit*. Forward rendering has used the light’s view this way for decades, in shadow maps [16, 42, 56] and translucent [9] and reflective [10] shadow maps, with hand-designed quantities on known geometry; we bring it into inverse rendering, where the geometry and the stored quantity are both learned.

Our method, LiSA (Light-Space Atlas), splats for every light the first two depth moments of the deposit and a few learned flux features, and filters them into a pyramid (Fig. 1). Any point at which shading is evaluated, a *receiver*, projects into the light and reads the pyramid: a Gaussian center while the geometry is optimized, a pixel afterwards. The moments yield a filtered shadow test in the spirit of variance shadow maps [16], refined by a residual network, and the flux features feed a transfer term that is linear in the deposited features, a learned relative of translucent shadow maps. The atlas costs one 512^2 rasterization per image, its inputs are relative to the light, and gradients flow through it into the Gaussians.

Training this model with a loss on display-encoded radiance collapsed thin, cluttered structures such as the bucket of the Lego bulldozer into blobs (Fig. 4). We therefore stage the optimization radiometrically: a network-free point-light model first fits the geometry while the target is annealed from display to linear radiance, as GS^3 and SSD-GS do for HDR data [5, 66]; the full model then trains in linear radiance on Gaussian receivers; and a final pass refines appearance on pixel receivers with the geometry frozen. In a controlled study on two scenes, changing only the loss domain raises held-out PSNR from 23.0 to 30.2 dB on Lego and from 24.1 to 30.1 dB on Drums.

On all 18 scenes of three public OLAT benchmarks [5, 15, 62], with GS^3 , SSD-GS, OLAT Gaussians [33] and RNG retrained under one protocol and scored against identical targets, LiSA attains the best average PSNR, SSIM and LPIPS (33.47 vs. 31.59 dB for the next best method) and trains in about 20 minutes per scene. Per benchmark, it leads the strongest baseline by 2.0 dB on the real captures (0.7 dB after per-view 2D alignment) and by 0.4 dB on the GS^3 scenes, and trails OLAT Gaussians by 0.1 dB on SSS-GS, making it the only method among the top two on every benchmark. Our ablations also show the limits of the transfer term: it matters most on translucent materials, but a local network of equal size matches it on most others. We contribute:

- a light-space formulation of point-light visibility and transport for Gaussian splats, in which one rasterization from the light records the deposit, from which receivers derive a filtered, learnable shadow test and a transfer term linear in the deposited features;
- a radiometrically staged optimization that makes this model trainable on thin, cluttered geometry, with a controlled study showing that the loss domain decides whether such geometry forms;

- an analysis on 18 scenes showing where light-space transfer helps and where a local network suffices, and that the test lights of current OLAT benchmarks lie a median of only 1.1° from a training light, so these benchmarks barely test light extrapolation.

2. Related Work

Relighting from point-light captures. Neural reflectance fields recover relightable volumes from OLAT illumination [3, 4], NRHints [62] conditions a radiance field on ray-traced shadow and highlight hints, and inverse rendering under environment light factors scenes into geometry and BRDFs [7, 20, 21, 27, 29, 34, 50, 65]. For point lights, GS^3 [5] combines spatial and angular Gaussians with splatted shadows and a residual network, RNG [17] shades neural Gaussian features with a network-refined depth shadow-map cue and switches from forward to deferred shading during training, OLAT Gaussians [33] are guided by a proxy mesh from neural surface reconstruction [53], BiGS [35] uses bidirectional spherical harmonics, SSS-GS [15] adds a neural subsurface residual, and SSD-GS [66] separates scattering, shadows and reflection. RadiosityGS [26] and stochastic ray tracing [57] solve light transport explicitly on Gaussians, neural transmittance and radiance transfer fields learn visibility and transfer for radiance fields [37, 48], PRTGS [22] precomputes radiance transfer for Gaussians under distant light, and relightable avatars learn shadows and transfer on a body model [6, 46, 54]. All of these evaluate visibility or scattering per primitive or per ray; LiSA moves both into the light’s image plane.

Visibility for Gaussians. Per-Gaussian visibility comes from splatting toward the light [5, 66], from tracing rays through Gaussians [20, 21, 40, 57], or from depth shadow maps [17], and deep Gaussian shadow maps [39] tabulate the transmittance of fixed Gaussians; our atlas is filtered, evaluated at arbitrary receivers, and learned with the geometry it describes.

Light-space rendering. Shadow maps [56] and their filtered variants [1, 2, 16, 18, 42, 45], shadow maps for semi-transparent occluders [24, 32, 36, 43, 47, 61], translucent shadow maps [9, 28] built on diffusion models [14, 25], reflective shadow maps [10, 11] and multi-scale light representations [8, 30] store per-light quantities and gather them at shading time, and precomputed radiance transfer makes outgoing light linear in the incident light [49]. Our shadow test is a two-moment test on an opacity-weighted deposit, close to variance and opacity shadow maps, and our transfer term resembles a translucent shadow map on pre-filtered levels without explicit sender positions. Learned counterparts filter shadow-map buffers for known geometry [12] or

decode rasterized buffers [19, 41, 52, 58, 64]; LiSA learns the stored quantity and the kernels while reconstructing the geometry.

The domain of the loss. RawNeRF [38] reweights a linear loss toward a tone-mapped error, and GS³ [5] anneals the gamma of its HDR targets toward linear radiance; in our model, the display domain prevented thin geometry from forming (Sec. 4).

3. Light-Space Atlases

We reconstruct an object from images $\{Y_m\}$, each taken by a calibrated camera under a point light at $\mathbf{p}_m$ with RGB intensity $\mathbf{I}_m$ (Fig. 2).

3.1. Preliminaries

The scene is a set of 3D Gaussians with means μ_i , covariances Σ_i and opacities o_i [31], each carrying a base color $\mathbf{b}_i \in \mathbb{R}^3$ and a latent material code $\mathbf{f}_i \in \mathbb{R}^{32}$ whose first three channels define a learned unit normal $\mathbf{n}_i$. Splatting composites per-Gaussian attributes $\mathbf{c}_i$ front to back,

$$\bar{\mathbf{c}}(u) = \sum_i w_i(u) \mathbf{c}_i, \quad w_i(u) = \alpha_i(u) \prod_{j < i} (1 - \alpha_j(u)), \quad (1)$$

where $\alpha_i(u)$ is the opacity of Gaussian i at pixel u . For a point $\mathbf{x}$ we write $\mathbf{l}, \mathbf{v}, \mathbf{h}$ for the unit light, view and half vectors and $\bar{E}(\mathbf{x}) = \mathbf{I}/\|\mathbf{p} - \mathbf{x}\|^2$ for the irradiance at normal incidence.

3.2. Splatting from the light records the deposit

Place a perspective camera at the light, aimed at the scene center, with a field of view that covers the splatted extent of 99.9% of the Gaussians. Its pixel u subtends a solid angle $d\omega_u$ and receives the flux $\mathbf{I} d\omega_u$. Splatting treats each Gaussian as absorbing the fraction $\alpha_i(u)$ of the light still travelling along the ray, so the flux it intercepts is

$$P_i(u) = \mathbf{I} d\omega_u w_i(u), \quad (2)$$

with the weights of Eq. (1). This holds within the rasterized splatting model, whose per-tile sorting by center depth, low-pass filter and clamped opacities approximate volumetric transmittance [23, 44, 51, 60]. No normal or distance enters Eq. (2): since an element dA at distance r and incidence θ receives $\mathbf{I} \cos \theta dA/r^2 = \mathbf{I} d\omega$, foreshortening and falloff are accounted for by how many light pixels a surface covers.

Atlas. Every Gaussian emits $\mathbf{c}_i = [\phi_i, z_i, z_i^2, 1]$, where z_i is the depth of its center along the light’s axis relative to the scene center, normalized by the scene radius, and $\phi_i = \text{softplus}(\text{MLP}_\phi(\mathbf{f}_i, \mathbf{b}_i, \mathbf{n}_i \cdot \mathbf{l}_i)) \in \mathbb{R}_+^C$ are $C = 8$

learned flux features. One rasterization at 512^2 yields the atlas

$$\mathbf{A}(u) = \sum_i w_i(u) \mathbf{c}_i = [\Phi(u), M_1(u), M_2(u), M_0(u)], \quad (3)$$

the deposit per unit of the pixel’s incident flux: M_0 is the fraction of that flux absorbed by the scene, M_1/M_0 and M_2/M_0 are the first two moments of the depth at which it is absorbed, and Φ describes what it hit. The absolute deposit $\mathbf{I} d\omega_u \mathbf{A}(u)$ also involves $d\omega_u \propto \cos^3$ of the off-axis angle, which varies by less than 30% across a typical atlas in our data (up to $2.1 \times$ at the corners of the widest one); the shadow statistics below are ratios that do not depend on it, and the transfer term ignores it. A pyramid $\mathbf{A}_0, \dots, \mathbf{A}_{K-1}$ with $K = 7$ follows from repeated binomial blurring and $2 \times$ decimation. Because every channel is linear in the weights w_i , each level still holds moments and feature sums of the deposit, over footprints growing from one texel to about a third of the object, which is what makes such shadow maps filterable [16, 42].

3.3. Shading from the atlas

A receiver $\mathbf{x}$ projects into the light view at $u(\mathbf{x})$ with normalized depth $z(\mathbf{x})$ and samples every level bilinearly. At level k we form the mean deposit depth $\bar{z}_k = M_{1,k}/M_{0,k}$, a spread $\eta_k = (M_{2,k}/M_{0,k} - \bar{z}_k^2 + \tau_k^2)^{1/2}$ that includes the texel size τ_k , the offset $\delta_k = z - \bar{z}_k$ of the receiver behind the deposit, and $t_k = \delta_k/\eta_k$. If the depth of the deposit within a footprint were normally distributed, the fraction of light absorbed in front of the receiver would be $M_{0,k} G(t_k)$ with the standard normal CDF G , which gives the moment test

$$V_k = 1 - M_{0,k} G(t_k - \beta), \quad (4)$$

where the bias $\beta = 3$, in units of the local spread, prevents self-shadowing. The descriptors $\mathbf{q}_k = [M_{0,k}, t_k, \delta_k, \log \eta_k, V_k]$ of all levels form $\mathbf{Q}(\mathbf{x}) \in \mathbb{R}^{5K}$. The test is a weak prior where an occluder and the receiver share a footprint (for a half-transparent occluder in front of an opaque receiver, $t_0 = 1$ and Eq. (4) returns 0.98 instead of 0.5, which the Chebyshev bound of variance shadow maps gets right [16]), and each Gaussian deposits the depth of its center, so a large, obliquely lit Gaussian appears at a single depth.

Visibility. A residual network $\mathcal{G}$, initialized to zero, corrects the finest test,

$$V(\mathbf{x}) = \sigma(\text{logit } V_0(\mathbf{x}) + \mathcal{G}(\mathbf{Q}, \mathbf{f}, \mathbf{n} \cdot \mathbf{l})), \quad (5)$$

so training starts from a classical filtered shadow map, and $\mathcal{G}$ learns from the multi-scale statistics to repair the cases above, the partial transmittance of fur and the soft shadow boundaries of real, finite lights.

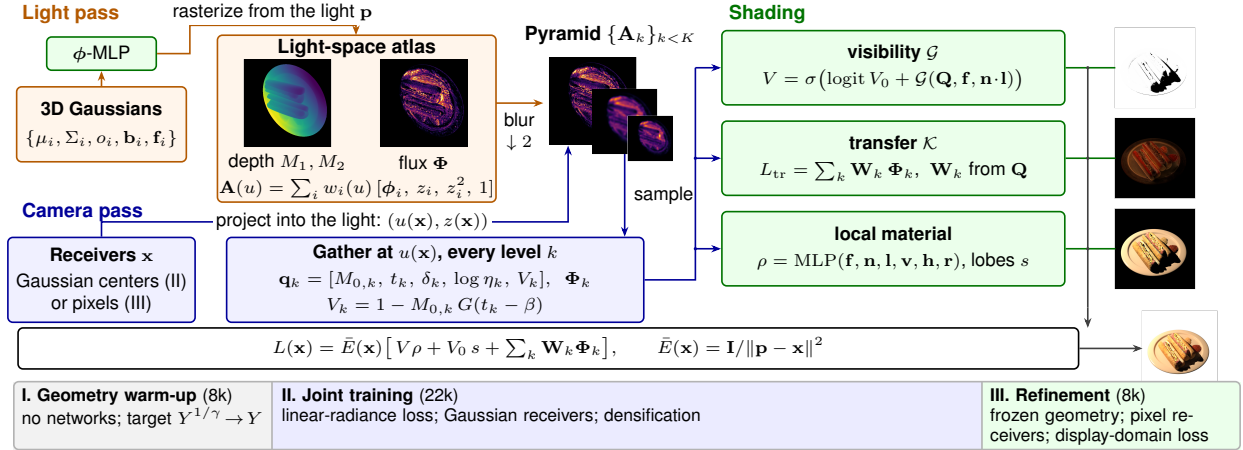


Figure 2. **Overview.** For every light, the Gaussians are rasterized once from the light; each deposits learned flux features ϕ_i and its depth moments with the compositing weight that equals the fraction of light it intercepts (Eq. (3)). A receiver projects into the light, gathers multi-scale statistics and flux from the pyramid, and is shaded by a residual on the moment shadow test, transfer kernels applied linearly to the gathered flux features, and a local material. Thumbnails: actual buffers of our Hotdog model on a test view. Bottom: the schedule of Sec. 4.

Transfer. By Eq. (2), light that enters a translucent object and leaves it at $\mathbf{x}$ contributes $\mathbf{I} \int \sum_i R(\mathbf{x}, \mu_i) w_i(\omega) d\omega$: a receiver-dependent kernel R integrated against the deposit over the light’s solid angle. Translucent shadow maps evaluate such integrals with stored irradiance and fixed diffusion kernels [9, 14, 28]. We learn both the deposited features and the kernels, using the pyramid levels as kernels of increasing width:

$$L_{\text{tr}}(\mathbf{x}) = \sum_k \mathbf{W}_k \Phi_k(\mathbf{x}), \quad (6)$$

$$\{\mathbf{W}_k\}_k = \text{softplus}(\mathcal{K}(\mathbf{Q}, \mathbf{f}, \mathbf{n} \cdot \mathbf{l}, \mathbf{n} \cdot \mathbf{v})), \quad \mathbf{W}_k \in \mathbb{R}_+^{3 \times C}.$$

The kernels depend only on the receiver and the geometric statistics $\mathbf{Q}$, so for a given geometry and light L_{tr} is linear in the deposited features; since the receiver’s own deposit lies in its footprint, nothing prevents L_{tr} from also absorbing part of the direct reflection, and the split between the terms is learned.

Local material and radiance. The local reflectance $\rho(\mathbf{x})$ is a latent-code neural BRDF: an MLP of $\mathbf{f}$, $\mathbf{n}$, Fourier encodings of $\mathbf{l}$, $\mathbf{v}$, $\mathbf{h}$ and the reflected view direction $\mathbf{r}$, their cosines and highlight hints $e^{\kappa(\mathbf{n} \cdot \mathbf{h} - 1)}$ [62], times a smoothed cosine $\psi(\mathbf{n} \cdot \mathbf{l}) = \text{softplus}(20 \mathbf{n} \cdot \mathbf{l})/20$. Position enters only through eight light-independent coefficients that mix eight light-dependent responses, which limits how much cast shadow can be baked into the material (the free per-Gaussian codes still allow some), and isotropic lobes $s(\mathbf{x})$ with fixed sharpness $\kappa \in \{8, \dots, 2048\}$ and light-independent weights carry sharp highlights. The outgoing radiance is

$$L(\mathbf{x}) = \bar{E}(\mathbf{x}) [V(\mathbf{x}) \rho(\mathbf{x}) + V_0(\mathbf{x}) s(\mathbf{x}) + L_{\text{tr}}(\mathbf{x})]. \quad (7)$$

The lobes use the geometric test V_0 , so highlights keep a gradient path where the learned V is still wrong. Factoring $\bar{E}(\mathbf{x})$ out lets the networks predict reflectance-like quantities; for L_{tr} it replaces the falloff at the entry points by that at the receiver, which is off by the squared ratio of their light distances, $(r_{\text{out}}/r_{\text{in}})^2$, for back-lit transmission; the kernels can absorb this only on average, which suffices when light distances vary little within a capture (constant in the synthetic scenes, within $1.8\times$ in the real ones; supplement). All inputs of $\mathcal{G}$ and $\mathcal{K}$ are relative to the light, so the operator is defined in the light’s frame rather than indexed by its position; whether this helps extrapolation is the subject of experiment E1 (Sec. 5.3).

3.4. Receivers: Gaussians or pixels

Gaussian receivers shade every Gaussian center μ_i and composite the colors with Eq. (1); *pixel receivers* composite $\mathbf{b}$, $\mathbf{f}$ and the expected depth first, lift each pixel to a world point and shade it once. Gaussian receivers give the geometry the ordinary rasterization gradients that densification relies on and keep overlapping surfaces from mixing materials, but they inherit the per-primitive shadow limitation of Sec. 1. Pixel receivers evaluate the shadow test at each pixel’s own surface point, so a shadow boundary can cross a large Gaussian. Like RNG’s switch from forward to deferred shading [17], LiSA uses Gaussian receivers while the geometry is optimized and pixel receivers once it is fixed (Sec. 4), with the same atlas.

4. Radiometrically Staged Optimization

4.1. Where the loss is computed

Photometric losses for relighting are commonly computed on display-encoded images, the native format of LDR captures. With the pure power law $\Gamma(L) = L^{1/\gamma}$, $\gamma = 2.2$, which we use to display HDR data and to linearize LDR data, the gradient of an ℓ_1 loss with respect to a predicted radiance $\hat{L}$ has magnitude

$$\Gamma'(\hat{L}) = \frac{1}{\gamma} \hat{L}^{1/\gamma-1}, \quad (8)$$

independently of the size of the residual. A pixel predicted at one percent of white thus receives twelve times the gradient of a white one, even when its residual is tiny. Since the prediction in Eq. (7) is a product of learned factors, a pixel is often dark early in training only because its visibility, normal or material is still wrong; we hypothesize that such pixels then dominate the gradients reaching the geometry, which the optimizer resolves by smoothing thin structures. The controlled study of Sec. 5.4 shows that the loss domain decides whether thin geometry forms in our model; it does not isolate the mechanism, and we have not tested remedies such as an offset inside Γ or loss reweighting [38].

4.2. Three stages

We seed 20k Gaussians uniformly inside the visual hull of 32 training silhouettes (a point is kept if it falls inside at least 90% of them) and initialize normals to point away from the scene center.

I. Geometry warm-up (8k iterations). A network-free point-light model colors every Gaussian with $\text{softplus}(\mathbf{b}_i) \bar{E}(\mu_i) \psi(\mathbf{n}_i \cdot \mathbf{l}_i)$, and only geometry, base colors and normals are optimized. The renderer output is linear while the target moves: it is the display-encoded image $\Gamma(Y)$ for the first two thirds of the warm-up, after which its exponent is annealed linearly from $1/\gamma$ to 1. The geometry thus forms against a target with compressed dynamic range and is carried continuously into linear radiance. GS³ and SSD-GS apply a comparable annealing to their HDR training images [5, 66]; we apply it to every data family and keep the joint training that follows in linear radiance.

II. Joint training (22k iterations). The full model of Eq. (7) is trained with Gaussian receivers and a linear-radiance loss; densification and pruning follow 3DGS [31] until iteration 25k, with at most 400k Gaussians.

III. Appearance refinement (8k iterations). Geometry and camera calibration are frozen, receivers switch to pixels, and the loss returns to the display domain in which images are evaluated. With the geometry fixed, the extra weight

on dark pixels improves all three metrics for both kinds of receivers in our experiments (Sec. 5.4).

All stages minimize $0.8 \ell_1 + 0.2 (1 - \text{SSIM})$, both computed in the current target domain, plus $0.5 \|\hat{\alpha} - \alpha\|_1$ on the rendered opacity with the foreground masks α provided by the benchmarks and a weak ℓ_2 penalty on the codes $\mathbf{f}$.

4.3. Implementation details

LiSA uses PyTorch and gsplat [59], an atlas of 512^2 texels with $K = 7$ levels and $C = 8$ flux channels, MLPs of width 128 (64 for visibility) initialized so that training starts from $L_{\text{tr}} \approx 0$ and $V = V_0$, and Adam (networks: 10^{-3} , decayed to about a quarter; Gaussians: as in 3DGS). On real captures we also refine the rotation of every training camera about its calibrated center and, after each step, remove the global translation that such refinements can absorb, so that the reconstruction stays in the calibrated frame; test cameras and lights are never adjusted. One configuration serves all 18 scenes, with only the data conventions of each benchmark differing, and a complete run takes 17 to 28 minutes (21 on average) on one NVIDIA RTX 6000 Ada.

5. Experiments

5.1. Setup

Data. We use every scene of three public OLAT benchmarks: the seven real captures of NRHints [62] (522–1500 training and 66–200 test images each), the six HDR synthetic scenes of GS³ [5] (2000/400), and the five subsurface-scattering scenes of SSS-GS [15] (500/500). We keep each benchmark’s conventions (512 px images, 256 px for SSS-GS; white background and $\gamma = 2.2$ display for HDR) and use all official training and all 5,691 test images.

Baselines. We compare with the four recent Gaussian methods that train per scene from point-light captures and provide code, GS³ [5], SSD-GS [66], OLAT Gaussians (OLAT-GS) [33] and RNG [17], retrained with their released code and default schedules (60k–100k iterations; ours: 38k). All use the foreground masks (to composite targets; OLAT-GS also for its surface stage, LiSA also for an opacity loss); GS³, SSD-GS and OLAT-GS refine training poses on real captures, and no method fits test cameras or lights. RNG and OLAT-GS fall below 21 dB on several scenes (*e.g.*, Drums, Lego), possibly because these violate their assumptions (black-background LDR data; a proxy surface); we report these runs unchanged. **[E5, E6] Add SSS-GS [15] once a stable reproduction exists (our runs diverged on three of five scenes) and reproduce one published configuration per baseline.**

Metrics and protocol. We report PSNR, SSIM [55] and VGG-LPIPS [63] on full images, averaged over the test

Table 1. **Comparison on all 18 scenes of three OLAT benchmarks** (5,691 test images, identical targets, original test calibration, no test-time fitting; best in **bold**, second underlined). Iterations: RNG 30k+70k, OLAT-GS 30k+30k, GS³ 100k, SSD-GS 100k (60k on SSS), LiSA 38k.

Method	NRHints, real (7)			GS ³ , synthetic (6)			SSS-GS, synthetic (5)			All scenes (18)		
	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓	PSNR↑	SSIM↑	LPIPS↓
RNG [17]	20.94	.828	.203	22.14	.895	.132	37.10	.983	.028	25.83	.893	.131
OLAT-GS [33]	27.26	<u>.886</u>	<u>.114</u>	27.11	.940	.079	40.74	.990	.016	30.95	.933	.075
GS ³ [5]	27.28	.884	.116	<u>31.97</u>	<u>.967</u>	<u>.049</u>	36.81	.983	.028	31.49	<u>.939</u>	.069
SSD-GS [66]	27.20	.882	.117	31.53	.965	.050	37.79	<u>.986</u>	<u>.023</u>	<u>31.59</u>	<u>.939</u>	<u>.068</u>
LiSA (ours)	29.28	.909	.100	32.40	.970	.038	<u>40.63</u>	.990	.016	33.47	.952	.056

views of each scene and then over scenes, and test per-scene differences with a Wilcoxon signed-rank test. All predictions are scored against one common set of 8-bit targets (changing the baselines’ averages by at most 0.005 dB), and foreground-masked PSNR preserves the ranking within every benchmark (supplement); main results use one run per method and scene, ablations three seeds. Earlier versions of LiSA were evaluated on these benchmarks during development (supplement); the final configuration was selected on views held out from the training sets and is shared by all scenes.

5.2. Comparison with the state of the art

LiSA has the best average PSNR, SSIM and LPIPS over the 18 scenes (Tab. 1), 1.89 dB ahead of SSD-GS, and its per-scene advantage over every baseline is significant ($p \leq 0.016$). No baseline is consistent across benchmarks: OLAT-GS is best on SSS-GS but trails LiSA by 5.3 dB on the GS³ scenes, and GS³ is second on its own scenes but last on SSS-GS. LiSA is the only method among the top two on every benchmark and has the best LPIPS on all three. The margin is concentrated, however: three real captures contribute 46% of it, LiSA and GS³ each win three of the six GS³ scenes in PSNR (LiSA wins five in LPIPS), and on the 11 synthetic scenes, where calibration is exact, LiSA leads SSD-GS by 1.77 dB ($p = 0.014$) but not OLAT-GS significantly ($p = 0.12$).

The largest gains occur where moving cast shadows and light crossing translucent objects dominate: 1.90 dB on Hot-dog and 1.91 dB on Translucent over the best baseline. In Fig. 3, the shadows of the vase handle and the hot dogs fall across large floor and plate Gaussians; per-primitive visibility fragments them, while pixel receivers reading the atlas reproduce their outlines. LiSA trails on Drums (−1.09 dB) and AnisoMetal (−0.85 dB), whose thin or anisotropic highlights our isotropic lobes miss.

On real captures LiSA leads by 2.0 dB on average, entirely due to Cat, Fish and Pixiu; on the other four it trails GS³ by 0.3 dB. Fixed-calibration scores here depend on registration as well as appearance: the official poses carry

small errors, and refining training poses without anchoring the reconstruction can shift it relative to the test cameras, which our gauge-constrained refinement (Sec. 4.3) avoids by construction. Aligning every test rendering by its best 2D shift (supplement) shows the effect: the baselines need 4.0–4.6 px against 1.7 px for LiSA, and the lead of LiSA shrinks to 0.7 dB, staying large on Cat and Pixiu but vanishing on Fish. **[TO RUN] E2: LiSA without pose refinement and a rotation-only test-pose alignment for all methods.**

5.3. What the atlas contributes

Table 2 ablates the atlas on a panel of five scenes covering opaque, metallic, furry, translucent and subsurface-scattering materials, including three of the scenes where we trail most; the full model varies by 0.11 dB across seeds. Using only the finest atlas level costs 0.44 dB and replacing the learned visibility by the moment test of Eq. (4) costs 0.24 dB; these single-seed differences only suggest that both matter, and the panel does not yet compare atlas visibility with per-primitive visibility or a plain shadow map at the same receivers, which E3 does.

Removing the transfer term costs 1.37 dB and raises LPIPS from 0.047 to 0.055, on all three seeds, and the cost follows the material (Tab. 3): it is negligible on Drums and FurScene, 1.42 dB on AnisoMetal, and 2.29 and 2.99 dB on Translucent and Soap. The model routes 6–7% of its linear radiance through the transfer term on the first three scenes but 17% on Translucent and 58% on Soap (over all 18 scenes, 6–9% on opaque synthetic objects and 20–58% on subsurface-scattering ones), a learned allocation rather than a physical decomposition.

To separate the atlas from the extra radiance path, we replace the atlas transfer by a capacity-matched MLP that reads the receiver’s code, position and light and view directions but not the atlas, keeping the atlas shadows. This local residual is on par with the full model on average (31.54 vs. 31.51 dB), so on most of these materials the transfer term mainly supplies an unshadowed, nonnegative radiance path that a local network provides equally well. The atlas is ahead only on Translucent, where light visibly crosses the object before

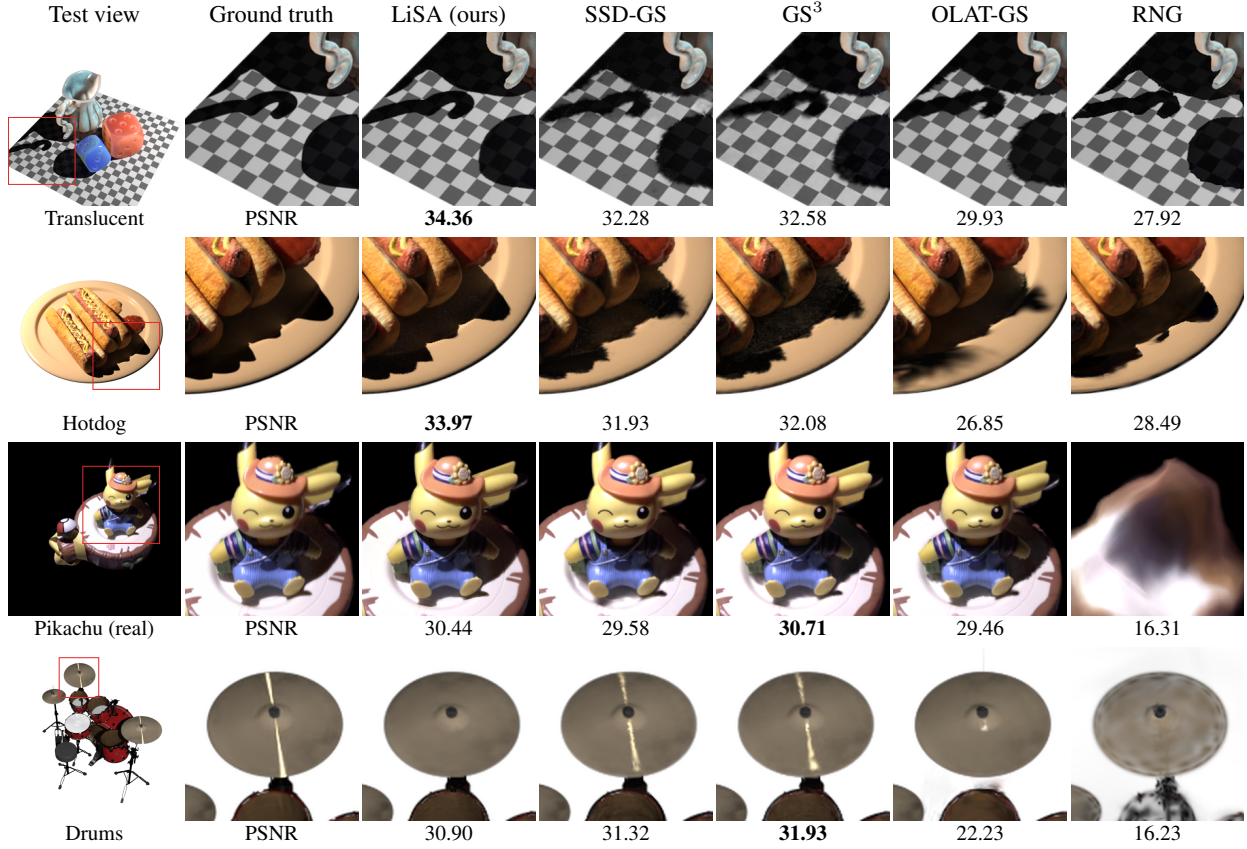


Figure 3. **Qualitative comparison** on test views (crops of the boxed region; PSNR of the full view), each at the median of LiSA’s per-view margin over the best baseline in its scene. Rows: the two GS³ scenes dominated by cast shadows, a real capture on which the methods are close, and the synthetic scene where we trail most. Per-primitive shadows of SSD-GS and GS³ fragment the moving shadows of the vase handle and the hot dogs, which pixel receivers reading the atlas reproduce; LiSA misses the thin streak on the Drums cymbal. More scenes, including Fish, are in the supplement.

Table 2. **Ablation of the atlas** on five scenes (Drums, AnisoMetal, Translucent, FurScene, Soap), scene-averaged. For rows with three seeds we report the mean over seeds and the standard deviation of PSNR; single-seed rows compare to the seed-0 full model (31.53 dB). [†]Capacity-matched MLP (49.1k vs. 49.6k parameters) that reads the receiver’s code, position and light/view directions but not the atlas; shadows still come from the atlas.

Variant	Seeds	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Full model	3	31.51 $\pm$ 0.11	.954	.047
w/o transfer term	3	30.15 $\pm$ 0.11	.948	.055
transfer $\rightarrow$ local residual [†]	3	31.54 $\pm$ 0.03	.954	.047
single atlas level ($K=1$)	1	31.09	.952	.049
moment test only ($g \equiv 0$)	1	31.29	.952	.048
per-Gaussian visibility [†]	3	TBD	TBD	TBD
hard shadow map [†]	3	TBD	TBD	TBD

[†][TO RUN] E3 (same receivers; atlas kept for transfer); also three seeds for the single-seed rows.

Table 3. **Where the transfer term matters.** Per-scene PSNR differences (mean $\pm$ std of paired differences over three seeds) of the full model against removing the transfer term and against the capacity-matched local residual, and the share of linear radiance the full model routes through the transfer term (mean over 8 test views).

Scene	vs. w/o transfer	vs. local residual	Transfer share
Drums	+0.10 $\pm$ 0.02	-0.09 $\pm$ 0.02	6%
FurScene	+0.02 $\pm$ 0.03	+0.02 $\pm$ 0.02	7%
AnisoMetal	+1.42 $\pm$ 0.69	-0.18 $\pm$ 0.75	7%
Translucent	+2.29 $\pm$ 0.23	+0.41 $\pm$ 0.26	17%
Soap	+2.99 $\pm$ 0.30	-0.29 $\pm$ 0.11	58%

reaching the visible surface (+0.41 $\pm$ 0.26 dB, positive for all three seeds but not significant), while the local residual is ahead on Soap (-0.29 $\pm$ 0.11 dB) and marginally on Drums; on AnisoMetal the full model is also less stable across seeds (26.4–27.9 dB). We therefore do not attribute our overall

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Table 4. **Held-out light directions ([TO RUN] E1)**. Mean PSNR over the five ablation scenes on the official test views and on training-split views whose lights fall in held-out 30° angular groups, after retraining every method without those views. [†]Three seeds.

Method	Official test	Held-out lights	Drop
GS ³	31.25	TBD	TBD
SSD-GS	31.01	TBD	TBD
OLAT-GS	28.67	TBD	TBD
LiSA, local residual [†]	31.54	TBD	TBD
LiSA [†]	31.51	TBD	TBD

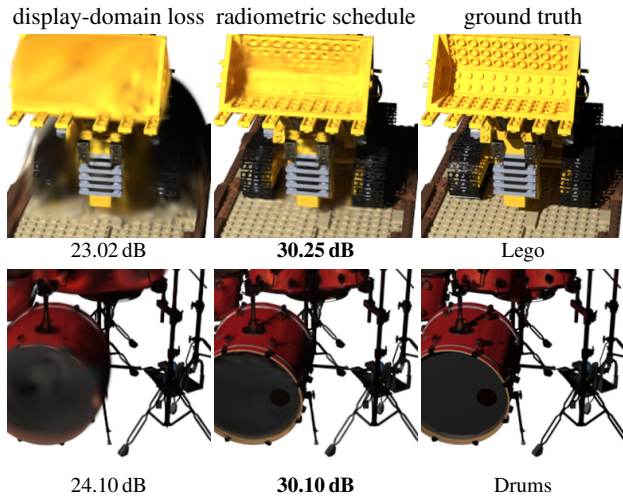


Figure 4. **Loss domain and geometry**. Two runs that share initialization, schedule, random state and every other setting, differing only in whether the loss is computed on display-encoded or (after the annealed warm-up) linear radiance; 16k iterations, PSNR over 200 held-out training views. All 200 views improve on both scenes.

margin to the transfer term, and hypothesize that it stems from the light-space visibility and the optimization schedule, which E3 and E4 test. The light-relative design should matter most under light extrapolation, which the benchmarks do not probe: their test lights lie a median of 1.1° from the nearest training light. **[TO RUN] E1 (Tab. 4): three seeds on the ablation panel; to be verified, not assumed: the atlas degrades less than the local residual.**

5.4. What the optimization contributes

Loss domain. On Lego and Drums, where our first runs failed, two 16k-iteration runs share the initialization, warm-up stage, schedule and random states, and differ only in the loss: the control uses display-encoded radiance throughout, ours follows stages I and II. On 200 views held out from the training set, the display-domain geometry collapses (Fig. 4), while the radiometric schedule raises PSNR from 23.02 to 30.25 dB on Lego and from 24.10 to 30.10 dB on Drums (LPIPS 0.122 $\rightarrow$ 0.057 and 0.060 $\rightarrow$ 0.034), improving

every view by a median of 7.2 and 6.0 dB. The study covers two HDR scenes, one seed and 16k iterations; on real and SSS-GS scenes, an earlier display-domain version of LiSA, which also differs otherwise, scores 0.2–0.4 dB higher (supplement). **[TO RUN] E4: on the ablation panel, three seeds: (a) display-domain loss throughout, (b) no warm-up, (c) pixel receivers in stage II, (d) no stage III, (e) an offset inside Γ .**

Receivers and refinement. On frozen 30k geometry of Lego, Drums and Soap (Tab. S6 in the supplement), refining in the display domain instead of linear radiance improves all three metrics for both kinds of receivers (PSNR +0.20 dB for Gaussians and +0.28 dB for pixels). Pixel receivers give the best SSIM and LPIPS (0.976 and 0.029) and Gaussian receivers a 0.19 dB higher PSNR; we keep pixels for their perceptual quality.

5.5. Cost

Profiled on an otherwise idle GPU on Drums, FurScene and Soap (Tab. S5 in the supplement), LiSA trains 1.6–5.5 times faster than GS³ and SSD-GS and uses 228k Gaussians on Drums against about 550k for both; over all 18 scenes, timed under less uniform conditions, its mean training time is 21 minutes against 55–107 for the baselines. The light pass keeps rendering interactive at 63–104 frames per second with a new light every frame, but makes LiSA slower than GS³ and SSD-GS on small scenes.

5.6. Limitations

Isotropic lobes miss brushed metal and thin streaks: on Drums, 0.27% of the pixels, flagged as narrow highlights in the ground truth, carry 72% of our excess squared error relative to GS³. The moment test is a weak prior for mixed footprints and center depths misplace large oblique Gaussians, for which Chebyshev or four-moment tests [16, 42] and intersection depths are natural fixes. One perspective atlas assumes the light lies outside the object, the transfer term ignores the solid-angle factor and the entry-point falloff, and light reaching a receiver only through regions the light cannot see is beyond one light-space gather.

6. Conclusion

Rasterizing Gaussians from a point light records the fraction of flux each primitive intercepts, and LiSA turns this deposit into a learned atlas that gives every receiver a filtered, refinable shadow test and a transfer term, for one extra rasterization per light. With a radiometrically staged optimization, it is the most consistent method across three benchmarks at a fraction of the usual training time; whether its light-relative design pays off for unseen light directions is the question our pending experiments address.

References

- [1] Thomas Annen, Tom Mertens, Philippe Bekaert, Hans-Peter Seidel, and Jan Kautz. Convolution shadow maps. In *Eurographics Symposium on Rendering (EGSR)*, 2007. 2
- [2] Thomas Annen, Tom Mertens, Hans-Peter Seidel, Eddy Flerackers, and Jan Kautz. Exponential shadow maps. In *Proceedings of Graphics Interface*, 2008. 2
- [3] Sai Bi, Zexiang Xu, Pratul Srinivasan, Ben Mildenhall, Kalyan Sunkavalli, Miloš Hašan, Yannick Hold-Geoffroy, David Kriegman, and Ravi Ramamoorthi. Neural reflectance fields for appearance acquisition. *arXiv preprint arXiv:2008.03824*, 2020. 1, 2
- [4] Sai Bi, Zexiang Xu, Kalyan Sunkavalli, Miloš Hašan, Yannick Hold-Geoffroy, David Kriegman, and Ravi Ramamoorthi. Deep reflectance volumes: Relightable reconstructions from multi-view photometric images. In *European Conference on Computer Vision (ECCV)*, 2020. 2
- [5] Zoubin Bi, Yixin Zeng, Chong Zeng, Fan Pei, Xiang Feng, Kun Zhou, and Hongzhi Wu. GS³: Efficient relighting with triple Gaussian splatting. In *SIGGRAPH Asia Conference Papers*, 2024. 1, 2, 3, 5, 6
- [6] Luis Bolanos, Shih-Yang Su, and Helge Rhodin. Gaussian shadow casting for neural characters. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [7] Hongze Chen, Zehong Lin, and Jun Zhang. GI-GS: Global illumination decomposition on Gaussian splatting for inverse rendering. In *International Conference on Learning Representations (ICLR)*, 2025. 2
- [8] Cyril Crassin, Fabrice Neyret, Miguel Sainz, Simon Green, and Elmar Eisemann. Interactive indirect illumination using voxel cone tracing. *Computer Graphics Forum*, 30(7), 2011. 2
- [9] Carsten Dachsbacher and Marc Stamminger. Translucent shadow maps. In *Eurographics Symposium on Rendering (EGSR)*, 2003. 2, 4
- [10] Carsten Dachsbacher and Marc Stamminger. Reflective shadow maps. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2005. 2
- [11] Carsten Dachsbacher and Marc Stamminger. Splatting indirect illumination. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2006. 2
- [12] Sayantan Datta, Derek Nowrouzezahrai, Christoph Schied, and Zhao Dong. Neural shadow mapping. In *ACM SIGGRAPH Conference Proceedings*, 2022. 2
- [13] Paul Debevec, Tim Hawkins, Chris Tchou, Haarm-Pieter Duiker, Westley Sarokin, and Mark Sagar. Acquiring the reflectance field of a human face. In *Proceedings of SIGGRAPH*, pages 145–156, 2000. 1
- [14] Eugene d'Eon, David Luebke, and Eric Enderton. Efficient rendering of human skin. In *Eurographics Symposium on Rendering (EGSR)*, 2007. 2, 4
- [15] Jan-Niklas DhiImann, Arjun Majumdar, Andreas Engelhardt, Raphael Braun, and Hendrik P. A. Lensch. Subsurface scattering for Gaussian splatting. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 121765–121789, 2024. 1, 2, 5
- [16] William Donnelly and Andrew Lauritzen. Variance shadow maps. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2006. 2, 3, 8
- [17] Jiahui Fan, Fujun Luan, Jian Yang, Miloš Hašan, and Beibei Wang. RNG: Relightable neural Gaussians. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 1, 2, 4, 5, 6
- [18] Randima Fernando. Percentage-closer soft shadows. In *ACM SIGGRAPH Sketches*, 2005. 2
- [19] Duan Gao, Guojun Chen, Yue Dong, Pieter Peers, Kun Xu, and Xin Tong. Deferred neural lighting: Free-viewpoint relighting from unstructured photographs. *ACM Transactions on Graphics*, 39(6), 2020. 3
- [20] Jian Gao, Chun Gu, Youtian Lin, Zhihao Li, Hao Zhu, Xun Cao, Li Zhang, and Yao Yao. Relightable 3D Gaussians: Realistic point cloud relighting with BRDF decomposition and ray tracing. In *European Conference on Computer Vision (ECCV)*, 2024. 2
- [21] Chun Gu, Xiaofei Wei, Zixuan Zeng, Yuxuan Yao, and Li Zhang. IRGS: Inter-reflective Gaussian splatting with 2D Gaussian ray tracing. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 2
- [22] Yijia Guo, Yuanxi Bai, Liwen Hu, Ziyi Guo, Mianzhi Liu, Yu Cai, Tiejun Huang, and Lei Ma. PRTGS: Precomputed radiance transfer of Gaussian splats for real-time high-quality relighting. In *Proceedings of the ACM International Conference on Multimedia (MM)*, 2024. 2
- [23] Letian Huang, Jiayang Bai, Jie Guo, Yuanqi Li, and Yanwen Guo. On the error analysis of 3D Gaussian splatting and an optimal projection strategy. In *European Conference on Computer Vision (ECCV)*, 2024. 3
- [24] Jon Jansen and Louis Bavoil. Fourier opacity mapping. In *Proceedings of the ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)*, pages 165–172, 2010. 2
- [25] Henrik Wann Jensen, Stephen R. Marschner, Marc Levoy, and Pat Hanrahan. A practical model for subsurface light transport. In *Proceedings of SIGGRAPH*, 2001. 2
- [26] Kaiwen Jiang, Jia-Mu Sun, Zilu Li, Dan Wang, Tzu-Mao Li, and Ravi Ramamoorthi. Differentiable light transport with Gaussian surfels via adapted radiosity for efficient relighting and geometry reconstruction. *ACM Transactions on Graphics (Proc. SIGGRAPH Asia)*, 2025. 2
- [27] Yingwenqi Jiang, Jiadong Tu, Yuan Liu, Xifeng Gao, Xiaoxiao Long, Wenping Wang, and Yuexin Ma. GaussianShader: 3D Gaussian splatting with shading functions for reflective surfaces. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [28] Jorge Jimenez, David Whelan, Veronica Sundstedt, and Diego Gutierrez. Real-time realistic skin translucency. *IEEE Computer Graphics and Applications*, 30(4), 2010. 2, 4
- [29] Haian Jin, Isabella Liu, Peijia Xu, Xiaoshuai Zhang, Songfang Han, Sai Bi, Xiaowei Zhou, Zexiang Xu, and Hao Su. TensolIR: Tensorial inverse rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. 2

- [30] Anton Kaplanyan and Carsten Dachsbacher. Cascaded light propagation volumes for real-time indirect illumination. In *Proceedings of the Symposium on Interactive 3D Graphics and Games (I3D)*, 2010. 2
- [31] Bernhard Kerbl, Georgios Kopanas, Thomas Leimkühler, and George Drettakis. 3D gaussian splatting for real-time radiance field rendering. *ACM Transactions on Graphics*, 42(4), 2023. 1, 3, 5
- [32] Tae-Yong Kim and Ulrich Neumann. Opacity shadow maps. In *Eurographics Workshop on Rendering Techniques*, 2001. 2
- [33] Zhiyi Kuang, Yanchao Yang, Siyan Dong, Jiayue Ma, Hongbo Fu, and Youyi Zheng. OLAT Gaussians for generic relightable appearance acquisition. In *SIGGRAPH Asia Conference Papers*, pages 1–11, 2024. 1, 2, 5, 6
- [34] Zhihao Liang, Qi Zhang, Ying Feng, Ying Shan, and Kui Jia. GS-IR: 3D Gaussian splatting for inverse rendering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [35] Zhenyuan Liu, Yu Guo, Xinyuan Li, Bernd Bickel, and Ran Zhang. BiGS: Bidirectional Gaussian primitives for relightable 3D Gaussian splatting. In *International Conference on 3D Vision (3DV)*, 2025. 1, 2
- [36] Tom Lokovic and Eric Veach. Deep shadow maps. In *Proceedings of SIGGRAPH*, 2000. 2
- [37] Linjie Lyu, Ayush Tewari, Thomas Leimkühler, Marc Habermann, and Christian Theobalt. Neural radiance transfer fields for relightable novel-view synthesis with global illumination. In *European Conference on Computer Vision (ECCV)*, 2022. 2
- [38] Ben Mildenhall, Peter Hedman, Ricardo Martin-Brualla, Pratul P. Srinivasan, and Jonathan T. Barron. NeRF in the dark: High dynamic range view synthesis from noisy raw images. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022. 3, 5
- [39] Aymen Mir, Riza Alp Guler, Jian Wang, Gerard Pons-Moll, and Bing Zhou. Animated 3DGS avatars in diverse scenes with consistent lighting and shadows. *arXiv preprint arXiv:2601.01660*, 2026. 2
- [40] Nicolas Moenne-Loccoz, Ashkan Mirzaei, Or Perel, Riccardo de Lutio, Janick Martinez Esturo, Gavriel State, Sanja Fidler, Nicholas Sharp, and Zan Gojcic. 3D Gaussian ray tracing: Fast tracing of particle scenes. *ACM Transactions on Graphics*, 43(6), 2024. 2
- [41] Oliver Nalbach, Elena Arabadzhiyska, Dushyant Mehta, Hans-Peter Seidel, and Tobias Ritschel. Deep shading: Convolutional neural networks for screen space shading. *Computer Graphics Forum*, 36(4), 2017. 3
- [42] Christoph Peters and Reinhard Klein. Moment shadow mapping. In *Proceedings of the ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)*, 2015. 2, 3, 8
- [43] Christoph Peters, Cedrick Münstermann, Nico Wetzstein, and Reinhard Klein. Beyond hard shadows: Moment shadow maps for single scattering, soft shadows and translucent occluders. In *Proceedings of the ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games (I3D)*, 2016. 2
- [44] Lukas Radl, Michael Steiner, Mathias Parger, Alexander Weinrauch, Bernhard Kerbl, and Markus Steinberger. StopThePop: Sorted Gaussian splatting for view-consistent real-time rendering. *ACM Transactions on Graphics*, 43(4), 2024. 3
- [45] William T. Reeves, David H. Salesin, and Robert L. Cook. Rendering antialiased shadows with depth maps. In *Proceedings of SIGGRAPH*, pages 283–291, 1987. 2
- [46] Shunsuke Saito, Gabriel Schwartz, Tomas Simon, Junxuan Li, and Giljoo Nam. Relightable Gaussian codec avatars. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 2
- [47] Marco Salvi, Kiril Vidimče, Andrew Lauritzen, and Aaron Lefohn. Adaptive volumetric shadow maps. *Computer Graphics Forum*, 29(4):1289–1296, 2010. 2
- [48] Mohammad Shafiei, Sai Bi, Zhengqin Li, Aidas Liaudanskas, Rodrigo Ortiz-Cayon, and Ravi Ramamoorthi. Learning neural transmittance for efficient rendering of reflectance fields. In *British Machine Vision Conference (BMVC)*, 2021. 2
- [49] Peter-Pike Sloan, Jan Kautz, and John Snyder. Precomputed radiance transfer for real-time rendering in dynamic, low-frequency lighting environments. *ACM Transactions on Graphics*, 21(3), 2002. 2
- [50] Pratul P. Srinivasan, Boyang Deng, Xiuming Zhang, Matthew Tancik, Ben Mildenhall, and Jonathan T. Barron. NeRV: Neural reflectance and visibility fields for relighting and view synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021. 2
- [51] Chinmay Talegaonkar, Yash Belhe, Ravi Ramamoorthi, and Nicholas Antipa. Volumetrically consistent 3D Gaussian rasterization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. 3
- [52] Justus Thies, Michael Zollhöfer, and Matthias Nießner. Deferred neural rendering: Image synthesis using neural textures. *ACM Transactions on Graphics*, 38(4), 2019. 3
- [53] Peng Wang, Lingjie Liu, Yuan Liu, Christian Theobalt, Taku Komura, and Wenping Wang. NeuS: Learning neural implicit surfaces by volume rendering for multi-view reconstruction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021. 2
- [54] Shaofei Wang, Tomas Simon, Igor Santesteban, Timur Bagautdinov, Junxuan Li, Vasu Agrawal, Fabian Prada, Shou-I Yu, Pace Nalbony, Matt Gramlich, Roman Lubachevsky, Chenglei Wu, Javier Romero, Jason Saragih, Michael Zollhoefer, Andreas Geiger, Siyu Tang, and Shunsuke Saito. Relightable full-body Gaussian codec avatars. In *ACM SIGGRAPH Conference Papers*, 2025. 2
- [55] Zhou Wang, Alan C. Bovik, Hamid R. Sheikh, and Eero P. Simoncelli. Image quality assessment: From error visibility to structural similarity. *IEEE Transactions on Image Processing*, 13(4):600–612, 2004. 5
- [56] Lance Williams. Casting curved shadows on curved surfaces. In *Proceedings of SIGGRAPH*, pages 270–274, 1978. 2
- [57] Peiyu Xu, Xin Sun, Krishna Mullia, Raymond Fei, Iliyan Georgiev, and Shuang Zhao. Stochastic ray tracing for the reconstruction of 3D Gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026. 2

- [58] Keyang Ye, Qiming Hou, and Kun Zhou. 3D Gaussian splatting with deferred reflection. In *ACM SIGGRAPH Conference Papers*, 2024. 3
- [59] Vickie Ye, Ruilong Li, Justin Kerr, Matias Turkulainen, Brent Yi, Zhuoyang Pan, Otto Seiskari, Jianbo Ye, Jeffrey Hu, Matthew Tancik, and Angjoo Kanazawa. gsplat: An open-source library for Gaussian splatting. *Journal of Machine Learning Research*, 26(34):1–17, 2025. 5
- [60] Zehao Yu, Anpei Chen, Binbin Huang, Torsten Sattler, and Andreas Geiger. Mip-splatting: Alias-free 3D Gaussian splatting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024. 3
- [61] Cem Yuksel and John Keyser. Deep opacity maps. *Computer Graphics Forum*, 27(2), 2008. 2
- [62] Chong Zeng, Guojun Chen, Yue Dong, Pieter Peers, Hongzhi Wu, and Xin Tong. Relighting neural radiance fields with shadow and highlight hints. In *ACM SIGGRAPH Conference Proceedings*, 2023. 1, 2, 4, 5
- [63] Richard Zhang, Phillip Isola, Alexei A. Efros, Eli Shechtman, and Oliver Wang. The unreasonable effectiveness of deep features as a perceptual metric. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018. 5
- [64] Xiuming Zhang, Sean Fanello, Yun-Ta Tsai, Tiancheng Sun, Tianfan Xue, Rohit Pandey, Sergio Orts-Escolano, Philip Davidson, Christoph Rhemann, Paul Debevec, Jonathan T. Barron, Ravi Ramamoorthi, and William T. Freeman. Neural light transport for relighting and view synthesis. *ACM Transactions on Graphics*, 40(1), 2021. 3
- [65] Xiuming Zhang, Pratul P. Srinivasan, Boyang Deng, Paul Debevec, William T. Freeman, and Jonathan T. Barron. NeRFactor: Neural factorization of shape and reflectance under an unknown illumination. *ACM Transactions on Graphics*, 40(6), 2021. 2
- [66] Iris Zheng, Guojun Tang, Alexander Doronin, Paul Teal, and Fang-Lue Zhang. SSD-GS: Scattering and shadow decomposition for relightable 3D Gaussian splatting. In *International Conference on Learning Representations (ICLR)*, 2026. 1, 2, 5, 6